GOLD-HAM: On-Policy Distillation for Selective Storage in Hybrid Associative Memory Architectures

Liu Xiao
liu.xiao.in@gmail.com

Abstract

As large language models (LLMs) transition from pre-training scale to inference-time reasoning scale, "System 2" Test-Time Compute (TTC) has become the primary lever for frontier performance. Knowledge distillation must transition in tandem: shifting from compressing static facts to transferring complex reasoning traces into efficient architectures. On-Policy Distillation (OPD) addresses the exposure bias of offline methods by having the student generate its own trajectories, while cross-tokenizer techniques like GOLD enable distillation across fundamentally different model families. In this paper we survey the state of the art and propose **GOLD-HAM-TTT**, a framework integrating on-policy distillation with Hybrid Associative Memory architectures and Test-Time Training (TTT). Rather than using static exact-match KV caches, our approach leverages the token-level OPD distillation signal to supervise In-Place Test-Time Training of continuous "fast weights" during inference. Furthermore, by utilizing an RWKV-7 linear backbone, we formalize sequence generation as an optimal control problem, allowing for Finite-Horizon Linear Quadratic Regulator (LQG/LQR) Test-Time Control. This unifies subquadratic sequence processing with genuine System 2 lookahead planning.

1 Introduction

The traditional LLM paradigm historically relied on scaling up pre-training clusters and vast, unstructured datasets. However, early 2026 marked a pivotal shift toward "System 2" reasoning paradigms, where models utilize Test-Time Compute (TTC) to "think" via internal monologues, tree-search, and backtracking before generating final answers. This transition dictates a fundamental evolution in post-training: we no longer merely distill knowledge; we distill *the reasoning process itself*.

However, traditional offline distillation suffers from exposure bias. If a student model is trained solely via teacher-forcing on pristine "System 2" reasoning traces, it will inevitably drift when generating its own multi-step thoughts during deployment. On-Policy Distillation (OPD) mitigates this by allowing the student to explore its own generation trajectories, explicitly regularizing its rollout against the teacher's sequence distribution.

In this survey, we explore the latest 2025–2026 developments in on-policy distillation, with a specific focus on cross-vocabulary methodologies such as Universal Logit Distillation (ULD) and General On-Policy Logit Distillation (GOLD). We also examine the broader landscape of On-Policy Self-Distillation (OPSD), where models leverage privileged information to act as their own teachers.

2 The Challenge of Cross-Tokenizer Distillation

When transferring knowledge between models with different tokenizers (e.g., distilling a Llama-3.1-8B-Instruct teacher into a SmolLM2-135M-Instruct student), two fundamental misalignment problems arise:

- **Sequence Misalignment:** A given string of text is tokenized into different sequence lengths by the teacher and the student. Token-by-token comparison of forward passes is impossible.
- **Vocabulary Misalignment:** The vocabulary spaces Ω^S and Ω^T differ entirely. Even if token lengths matched, identifying which logits in the teacher distribution correspond to which logits in the student distribution is non-trivial.

Early solutions, such as simple sequence truncation to the minimum length, resulted in unacceptable information loss, particularly for dense reasoning tasks.

3 General On-Policy Logit Distillation (GOLD)

Building upon the offline Universal Logit Distillation (Boizard et al., 2025), the GOLD framework, as popularized by the HuggingFace H4 team, provides a robust solution for on-policy distillation across any model family.

3.1 Sequence Alignment via Token Merging

Instead of truncation, GOLD leverages a token merging strategy rooted in the product rule of probability, $P(AB) = P(A)P(B|A)$. By identifying necessary merge operations to equalize sequence boundaries, GOLD sums the log-probabilities of constituent sub-tokens. This allows the framework to align differing token sequences without losing semantic content from the trajectory.

3.2 Vocabulary Alignment via Hybrid Loss

To address vocabulary disparities, GOLD introduces a hybrid mapping approach:

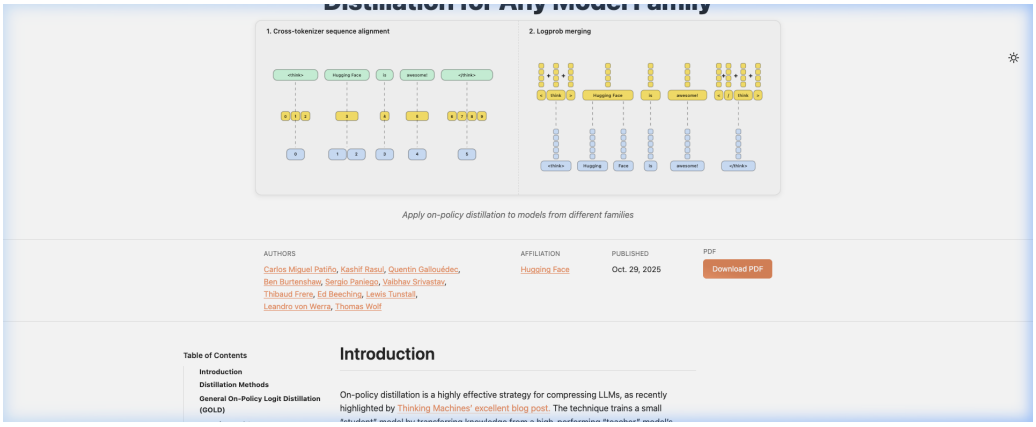

Figure 1: Interactive demonstration of Cross-Tokenizer Sequence Alignment and Logprob Merging in GOLD.

1. **Mapped Tokens:** For tokens that share an exact string match between the teacher and student vocabularies, a standard direct logit mapping is established. For these tokens, traditional KL Divergence (the core GKD loss) is applied.

2. **Unknown Tokens:** For tokens without a direct string-to-string mapping, GOLD falls back to an Optimal Transport approach (using Wasserstein distance). By sorting and padding softmax vectors, it treats the unmapped tokens as a permutation-invariant distribution, effectively merging probabilities across differing subword representations.

Experiments demonstrate that GOLD significantly outperforms offline ULD and traditional Reinforcement Learning (RL) baselines like Group Relative Policy Optimization (GRPO) on complex, multi-step math reasoning datasets like GSM8K.

4 On-Policy Self-Distillation (OPSD) and Reasoning

By 2026, the cost of running massive external teacher models for dense, token-level supervision led to the rise of On-Policy Self-Distillation (OPSD). In this paradigm, a single model assumes both roles.

4.1 Privileged Information and Dual-Role Architectures

In Self-Distillation, the "teacher" is manifested by providing the model with privileged information—such as ground-truth reasoning traces or verifiable factual answers. The model generates "gold" evaluations of student trajectories. The student version of the model, which only sees the prompt, is then trained to minimize the divergence between its outputs and the privileged teacher's evaluations.

4.2 Mitigating Exposure Bias and Mode Collapse

OPSD effectively removes the dependency on an external API or massive parameter set. Recent f-divergence theoretical frameworks categorize these self-play methodologies as essential for scaling reasoning. Methods like Self-Play Fine-Tuning (SPIN) minimize reverse KL divergence on their own rollouts, pushing the model to clearly separate verified correct paths from its own hallucinated distractors.

However, OPSD introduces new risks, notably "information leakage" (where the model memorizes the structure of the privileged context rather than the underlying reasoning step) and "mode collapse" (an echo chamber effect where biases are amplified). To counteract this, state-of-the-art pipelines frequently anchor self-distilled signals alongside sparse, verifiable environmental rewards (e.g., Self-Distilled RLVR).

5 Empirical Failure Modes of On-Policy Distillation

While OPD resolves the exposure bias problem, recent extensive studies—most notably by Fu et al. (arXiv:2603.25562)—have demonstrated that token-level OPD presents unique fragilities, particularly in long-horizon reasoning.

5.1 Estimator Tradeoffs: Bias vs. Variance Bounds

From an estimator perspective, sampled-token OPD represents a single-step truncation of the full causal return-to-go form of the sequence-level reverse-KL divergence. Let $c_t = (x, y_{<t})$ denote the prefix context at step t. The full sequence-level estimator relies on the future reward coupling $r_t = \log \pi_\theta(y_t|c_t) - \log q(y_t|c_t)$.

By dropping the future-coupling sum to reduce variance, token-level OPD becomes biased relative to the sequence-level objective. However, this truncation offers extreme stability advantages. Theoretical bounding proofs show that the worst-case variance upper bound for token-level OPD scales at $\mathcal{O}(T^2)$, whereas the full sequence-level estimator variance scales quartically at $\mathcal{O}(T^4)$ across a sequence length T.

5.2 Three Fragilities of Token-Level OPD

Despite the favorable variance bound, the common practice of reducing distribution matching to a single-token sampled signal inherently leads to three primary failure modes:

1. **Imbalanced Learning Signal:** Single-token signals are high variance and suffer from representation imbalance across a rollout. Most sampled tokens are highly penalized while learning signals collapse onto a few locally favorable subset tokens.

2. **Unreliable Teacher Guidance:** As a student-generated prefix drifts further from traces the teacher normally visits, the teacher's condition logit distribution becomes

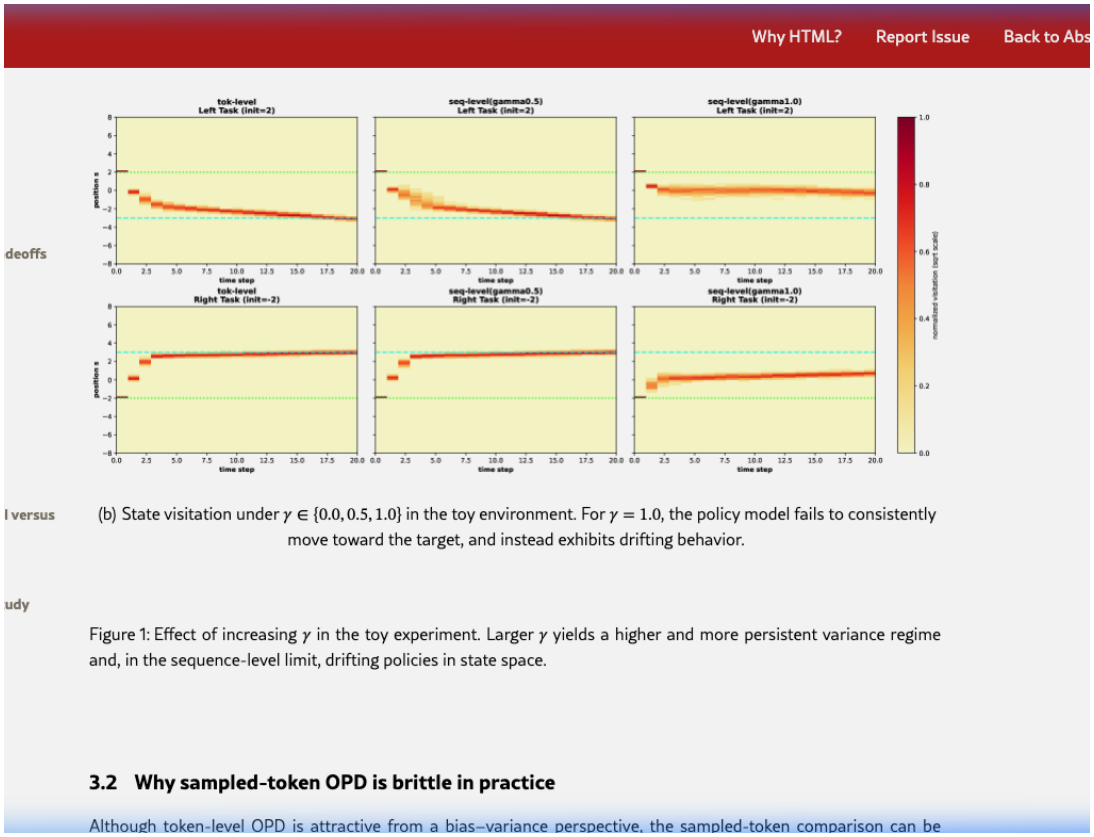

Figure 2: Empirical gradient variance and distribution mismatch as sequence-level reward coupling (γ) increases. The higher the variance, the higher the propensity for the model pathway to drift into unrecoverable failure modes (Fu et al., 2026).

sharply unreliable, providing erratic feedback and assigning high probabilities to mathematically unsound text.

3. **Tokenizer/Special-Token Distortion:** Extrinsic mismatches compounding over multi-step reasoning models cause catastrophic forgetting of the exact text formats required for verifiers.

5.3 Simple Fixes: Truncated Reverse-KL and Local Support Matching

To stabilize the optimization, Fu et al. propose moving away from strictly point-estimate OPD. The core methodological fix involves *teacher top-K local support matching*. At each prefix, the objective compares the teacher and student distributions solely over a teacher-defined locally plausible support set instead of rewarding only one sampled token.

Implemented mathematically as a truncated reverse-KL objective with top-p rollout sampling, this filters out the noisy, unreliable tail of the teacher's distribution. When combined with strict special-token masking to eliminate structural format drift, this corrected objective demonstrates far more stable learning curves and superior downstream metrics across both agentic and complex mathematical benchmarks.

6 Scaling with Multi-Domain Paradigms: Nemotron-Cascade 2

The transition from single-domain alignment to generalized, large-scale intelligence is well-exemplified by the newly released Nemotron-Cascade 2 model (2603.19220).

Nemotron-Cascade 2 represents a hyper-efficient 30B Mixture-of-Experts (MoE) architecture with only 3B activated parameters during inference. Remarkably, despite its compact active footprint, the model achieved a Gold Medal-level performance in the 2025 International Mathematical Olympiad (IMO) and ICPC World Finals—a feat traditionally reserved for foundation models completely surpassing the 600B parameter class, such as DeepSeekV3.2-Speciale.

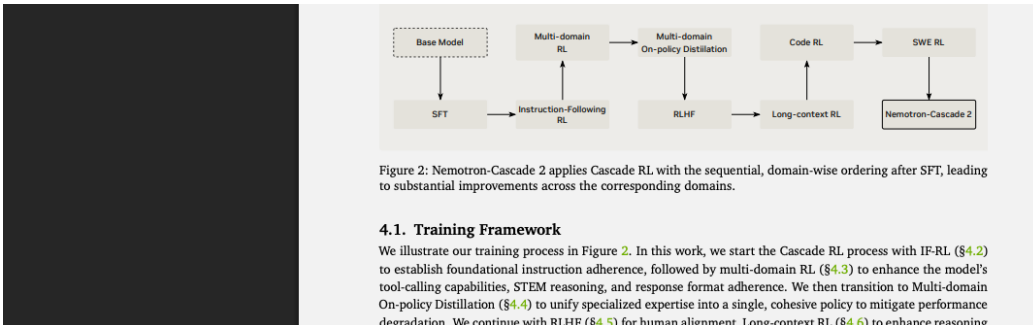

Figure 3: The Cascade RL process combined with Multi-Domain On-Policy Distillation utilized by Nemotron-Cascade 2 to scale high-tier reasoning without degrading alternative benchmarks.

To achieve this extreme intelligence density (a $\sim 20\times$ parameter reduction for IMO Gold attainment), the architecture institutes a *Cascade RL* pipeline tightly integrated with *multi-domain on-policy distillation.*

Rather than relying on a monolithic target capable of suffering generalized capability degradation, Nemotron sequentially cascades intermediate distillation updates drawn strictly from the strongest respective domain-specialized teachers at every milestone throughout its reinforcement learning cycles. This combination of structural self-rollouts driven by reinforcement learning, guided dynamically by overlapping on-policy distillation from domain experts, guarantees retention across domains. As a result, Nemotron-Cascade 2 cleanly

avoids the "tax" of benchmark regression while aggressively pushing the boundaries of agentic and mathematical capacity.

7 Cross-Architecture Distillation: From Transformers to RWKV and State-Space Models

The techniques surveyed thus far—GOLD, truncated reverse-KL, and multi-domain cascading—address the challenge of transferring knowledge between models that share the same fundamental architecture (Transformers) but differ in tokenizer or scale. An even more ambitious frontier is *cross-architecture distillation*: migrating the capabilities of quadratic-complexity Transformer teachers into fundamentally different, subquadratic student architectures such as RWKV and Mamba. This setting compounds every alignment challenge discussed previously, while introducing entirely new ones rooted in representational heterogeneity.

7.1 Motivation: Why Migrate Away from Transformers?

Despite their dominance in LLM capabilities, Transformers suffer from $\mathcal{O}(n^2)$ attention complexity and linearly growing KV-cache memory during inference. For deployment scenarios requiring long-context processing, real-time streaming, or edge inference, these costs become prohibitive. Two alternative architectures have emerged as compelling targets for distillation:

- **RWKV (Receptance Weighted Key Value):** A 100% RNN-based, attention-free architecture that achieves $\mathcal{O}(n)$ time complexity and $\mathcal{O}(1)$ memory during inference (no KV-cache). The latest iteration, RWKV-7 "Goose," replaces standard linear attention with a Generalized Delta Rule, enabling dynamic state evolution that surpasses the TC^0 expressive-power limitations of earlier linear-attention models.
- **Mamba (Selective State-Space Models):** An SSM-based architecture that uses selective scan mechanisms for context-dependent state updates, also achieving subquadratic complexity. Mamba-based models have demonstrated competitive performance with Transformers at equivalent scale when properly trained or distilled.

The key insight is that retraining these architectures from scratch on trillions of tokens is prohibitively expensive. Cross-architecture distillation offers a dramatically more efficient path: leverage the massive investment already embedded in pretrained Transformer checkpoints and *transfer* that knowledge into the more efficient architecture.

7.2 The Triple Alignment Challenge

Distilling a Transformer into RWKV or Mamba introduces a *triple alignment* problem that extends far beyond the cross-tokenizer setting:

1. **Tokenizer Mismatch (Vocabulary Alignment):** RWKV uses its own Trie-based "World Tokenizer" with a 65,536-token vocabulary, while typical Transformer teachers (Llama, Qwen) use BPE-based tokenizers with different vocabulary sizes and segmentation patterns. The GOLD-style hybrid loss and token merging techniques discussed in Section 3 become essential here, but the vocabulary overlap between a BPE tokenizer and a Trie tokenizer is typically *much lower* than between two BPE tokenizers, making the Wasserstein-distance fallback path far more heavily utilized.

2. **Sequence Mixing Mismatch (Architectural Alignment):** The self-attention mechanism in Transformers and the state-evolution mechanism in RWKV/Mamba perform fundamentally different sequence transformations. In attention, each token can directly attend to any previous token with learned, content-dependent weights. In RWKV-7, information flows through a compressed hidden state governed by the Generalized Delta Rule. Naive output-logit matching alone cannot bridge this representational gap—the intermediate representations encode information in structurally incompatible ways.

3. **Capacity Mismatch (Expressiveness Alignment):** Transformers can perform arbitrary pairwise token comparisons within a context window, while linear-time architectures must compress all history into a fixed-dimensional state. Certain capabilities—such as precise associative recall over very long contexts—may be fundamentally harder to represent in the student architecture, requiring the distillation process to gracefully handle *inexpressible* teacher behaviors rather than forcing catastrophic approximation.

7.3 Multi-Stage Distillation Frameworks

State-of-the-art cross-architecture distillation employs multi-stage pipelines to address each alignment layer progressively:

Stage 1: Attention-to-SSM Bridging. Lightweight MLP-based bridge modules are introduced to map the Transformer's Query/Key projection outputs into the RWKV or Mamba mixing space. This enables *layer-wise* supervision where the student's sequence mixing layers are trained to approximate the teacher's attention layer outputs, rather than relying solely on final logit matching. The MOHAWK framework formalizes this by viewing both attention and SSM mechanisms as abstract "sequence transformations," enabling granular layer-to-layer alignment.

Stage 2: Hidden State Alignment. Beyond the mixing layers, the student's hidden states at each layer are supervised to approximate the teacher's intermediate representations. This stage transfers the rich, distributed knowledge encoded in the Transformer's residual stream into the SSM's state dynamics. Principled weight initialization—projecting QKV weights from the Transformer into the RWKV/Mamba parameter space—has been shown to significantly accelerate convergence compared to random initialization.

Stage 3: On-Policy Logit Distillation. Finally, standard on-policy logit distillation (using GOLD or truncated reverse-KL) is applied at the output level, with the student generating its own rollouts. This stage is where all the cross-tokenizer techniques from Sections 3–5 become critical, as the student must learn to produce correct outputs *in its own tokenization space* while matching the teacher's distributional intent.

7.4 Hybrid Architectures and the Distillation Continuum

Recent practice has increasingly favored *hybrid* architectures that interleave Transformer attention layers with RWKV/Mamba layers, rather than attempting pure architectural migration. Models such as Jamba (Mamba-Transformer hybrid) and ARWKV (Attention-augmented RWKV) selectively retain a small number of attention layers for tasks requiring precise associative recall, while using linear-time layers for the majority of sequence processing.

From a distillation perspective, hybrids offer a natural *continuum*: one can progressively replace attention layers with SSM layers and distill the remaining knowledge at each step, rather than attempting a single catastrophic architectural leap. This "progressive hybridization" strategy has proven more robust than end-to-end cross-architecture distillation, particularly for reasoning-heavy tasks where the attention mechanism's exact-match capabilities are difficult to fully replicate in linear-time models.

7.5 The M1 Case Study: Reasoning via Distillation + RL

The M1 model provides a compelling case study of the full pipeline in action. Built on a Mamba-based (linear RNN) backbone, M1 achieves competitive reasoning performance through a three-phase process:

1. *Distillation from Strong Transformer Reasoners:* Initial knowledge transfer from state-of-the-art Transformer reasoning models using multi-stage hidden-state and logit alignment.

2. *Supervised Fine-Tuning (SFT):* Targeted training on curated reasoning datasets to refine the distilled representations.

3. *Reinforcement Learning (RL):* On-policy self-improvement using verifiable rewards (e.g., mathematical correctness), analogous to the self-distillation paradigm from Section 4 but applied to the subquadratic architecture.

This pipeline demonstrates that cross-architecture distillation is not a standalone technique but rather *the entry point* into a broader post-training workflow that includes all the on-policy and self-play methods surveyed in this paper.

8 Proposed Method: GOLD-HAM — Distillation for Selective Associative Memory in RWKV

The survey above reveals a critical gap: existing cross-architecture distillation methods are heavily focused on static layer replacement. However, a major recent breakthrough in sequence modeling is the **Hybrid Associative Memory (HAM)** layer (Lufkin et al., 2026). Rather than assigning entire layers to act as either self-attention or RNNs, the HAM architecture uses an intra-layer data-dependent routing mechanism. We propose **RWKV-HAM**, a novel architecture where an **RWKV-7** linear backbone (utilizing the Generalized Delta Rule) compresses predictable, continuous context into a fixed-size state, while an augmented exact-match KV cache selectively stores only "surprising" or high-value tokens that the RWKV state cannot easily predict.

The core challenge of this architecture is training the routing mechanism (typically a lightweight MLP) to accurately determine which tokens require high-resolution KV caching. We propose unifying on-policy cross-tokenizer distillation (GOLD) with the RWKV-HAM architecture by using the token-level OPD loss as the ground-truth signal for the selective storage router.

8.1 Key Insight: OPD Loss as a Selective Storage Signal

In a standard Transformer, every token is committed to the KV cache. In a HAM model, token x_t is only cached if the routing gate $g_\theta(x_t)$ exceeds a continuous threshold. We hypothesize that the on-policy reverse-KL divergence between the Transformer teacher and the HAM student on student-generated rollouts provides the optimal proxy for "surprisal."

Let $\mathcal{L}_t^{\text{OPD}}$ denote the token-level distillation loss at step t. If $\mathcal{L}_t^{\text{OPD}}$ is high, the RWKV-7 state has failed to adequately capture the necessary context to match the teacher's distribution, indicating that x_t contains complex associative information that must be explicitly cached. We formulate the router objective as a meta-learning loss:

$$\mathcal{L}_{\text{router}} = \sum_{t=1}^{T} \Big(g_\theta(x_t) - \sigma\big(\alpha \cdot \mathcal{L}_t^{\text{OPD}} - \beta\big) \Big)^2 + \lambda \sum_{t=1}^{T} g_\theta(x_t) \tag{1}$$

where σ is a sigmoid function, α scales the distillation loss, β represents the base predictability threshold, and λ acts as a sparsity penalty ensuring the KV cache does not grow blindly.

8.2 Three-Phase Training Pipeline

Phase 1: Full-Cache Warm-Up with GOLD. Begin with the HAM student architecture, but force the routing gate $g_\theta(x_t) \approx 1$ for all tokens, effectively mimicking a standard Transformer KV cache. Apply GOLD on-policy distillation from the Transformer teacher (e.g., Llama-3.1-8B) using the full cross-tokenizer alignment stack (token merging + hybrid optimal transport). This phase initializes the student's output projections without the added instability of dynamic cache sizing.

Phase 2: Distillation-Guided Router Training. Unfreeze the routing MLP and gradually increase the sparsity penalty λ. During student rollouts, compute the token-level OPD loss $\mathcal{L}_t^{\text{OPD}}$ against the teacher. Update the router to predict this loss using Equation 3. Simultaneously, the RWKV-7 state transitions are supervised by the teacher's hidden states via MOHAWK-style layer bridging. This phase forces the model to encode "easy" text structures into the RWKV recurrent state, reserving the KV cache strictly for tokens where the offline linear approximation degrades.

Phase 3: On-Policy Self-Refinement. Once the router threshold β stabilizes, the cache size will scale dynamically based purely on the complexity of the prompt (data-dependent growth). Apply on-policy self-distillation with verifiable rewards (e.g., RLVR on GSM8K). In this phase, the token-level supervision ensures the HAM model can perform extended chain-of-thought operations without suffering the linear memory explosion of pure Transformers, while maintaining the structural exactness required for final mathematical derivations.

8.3 Why On-Policy Matters More for Hybrid Associative Memories

The case for on-policy distillation is *stronger* in the HAM setting than in same-architecture distillation, for two structural reasons:

1. **State Compression Amplifies Exposure Bias:** The RWKV-7 component compresses history. If an offline-trained model makes a generative mistake during inference and fails to cache the erroneous token, the RWKV state will drift unrecoverably. On-policy training—where the student encounters its own errors—forces the router to recognize drifted configurations as "surprising" and aggressively cache correction tokens, preventing cascading failures specific to the linear recurrence.

2. **Tokenizer Misalignment Alters Information Density:** The token merging strategy in GOLD (summing log-probabilities of sub-tokens) changes the information density per time-step. A BPE token from the teacher might be split into three Trie tokens in the student. On-policy training allows the HAM router to dynamically calibrate its storage threshold across these differing subword segmentations, caching

only the heaviest sub-units rather than blindly following the teacher's sequence boundaries.

8.4 Expected Experimental Setup

We outline the planned experimental protocol:

- **Teacher:** Qwen2.5-7B-Instruct (Standard Transformer, 151K vocabulary)
- **Student:** HAM architecture (1.5B parameters), utilizing an RWKV-7 linear backbone coupled with an augmented selective KV cache (65K vocabulary)
- **Distillation:** GOLD with token merging + hybrid vocabulary loss mapped to the router objective
- **Baselines:** (a) Offline KD into pure RWKV-7, (b) Offline KD into HAM without selective routing (fixed 25% cache), (c) GOLD into same-architecture Transformer
- **Evaluation:** Memory reduction (GBs) vs. Pass@1 on reasoning benchmarks (GSM8K, MATH)

9 System 2 Distillation via In-Place Test-Time Training (TTT)

Extending the GOLD-HAM framework, we push beyond static latency optimizations into the realm of dynamic, associative inference. In late 2025 and 2026, researchers demonstrated that exact-match KV caching is computationally dead-ended for millions of tokens. The paradigm shifted toward **In-Place Test-Time Training (TTT)** [13], where the model weights themselves update dynamically during inference to memorize the prompt context.

9.1 Distillation as TTT Supervision

In a pure TTT-E2E model, "fast weights" are updated using generic self-supervised reconstruction objectives (e.g., predicting the next layer's activations). We propose **GOLD-HAM-TTT**: utilizing the token-level on-policy distillation loss to supervise continuous gradient descent on the HAM routing and projection layers *during student inference*.

Instead of routing a surprising token into an $\mathcal{O}(n)$ KV cache, the high OPD loss $\mathcal{L}_t^{\text{OPD}}$ triggers an instantaneous backpropagation step onto the mutant MLP parameter block corresponding to the current state. The student model literally "learns from the teacher's ghost" on the fly. This prevents context length from ballooning while retaining the exact associative mechanisms required for System 2 algorithmic operations (like long division or code execution), all handled within a constant $O(1)$ memory footprint.

10 Planning Before Prediction: Test-Time Control (TTC) in RWKV

Finally, distributing System 2 reasoning (distilled from deep search trees) into recurrent student networks presents a critical architectural barrier. A fundamental property of "System 2" thinking is lookahead planning and backtracking. Traditional Transformers achieve this sloppily via chain-of-thought tokens or explicit tree searches, which geometrically multiply computation.

10.1 Finite-Horizon LQR over Linear States

Because RWKV-7 operates as a continuous-time sequence model built on the Generalized Delta Rule [12], its hidden state h_t evolves according to a specific linear dynamical system geometry. We observe that this continuous numeric vector directly permits optimal control theory applications that categorical token streams reject.

We propose a **Test-Time Control (TTC)** framework for the RWKV-HAM architecture. Instead of generating text to "think," the model treats the generation process as a control problem to reach a target constraint surface (the correct answer manifold). We apply a Finite-Horizon Linear Quadratic Regulator (LQR) to the RWKV latent state equations:

$$x_{k+1} = A_k x_k + B_k u_k \quad \text{(RWKV-7 state evolution)} \tag{2}$$

$$J = \sum_{k=0}^{N-1} (x_k^\top Q x_k + u_k^\top R u_k) + x_N^\top Q_f x_N \quad \text{(LQR Cost Function)} \tag{3}$$

where u_k is the latent generative "action", and J represents the distance to the distilled reasoning goal. By calculating the optimal u_k sequences analytically over the linear recurrent state, the RWKV student can "plan" N steps into the future in continuous vector space before emitting a single text token. This effectively marries the subquadratic speed of linear RNNs with the deep System 2 planning previously reserved solely for massive, multi-agent reinforcement learning pipelines.

11 Conclusion

This paper surveys the rapidly evolving landscape of on-policy distillation—from cross-tokenizer alignment (GOLD) to failure-mode mitigation (truncated reverse-KL) and multi-domain scaling. Building on this established research, we address the pressing 2026 dilemma of reasoning-heavy System 2 architectures.

By unifying Hybrid Associative Memory models utilizing RWKV-7 backbones with In-Place Test-Time Training (TTT), we arrive at the **GOLD-HAM-TTT** framework. In this paradigm, on-policy distillation acts not merely as a post-training SFT replacement,

but as the dynamic signal that guides selective fast-weight updates during active inference. Finally, by treating the underlying RWKV-7 state algebra as a continuous dynamical system subject to Test-Time Control (TTC) LQR planning, we chart a rigorous mathematical path to embedding multi-step System 2 search capabilities directly into subquadratic hardware budgets. As artificial intelligence fundamentally shifts from pre-training memorization to inference-time planning, the fusion of continuous linear associative memory with on-policy generative alignment represents the frontier of efficient reasoning.

References

[1] Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the Knowledge in a Neural Network.* arXiv preprint arXiv:1503.02531.

[2] Schmidt, F. (2019). *Generalization in Generation: A closer look at Exposure Bias.* Proceedings of the 3rd Workshop on Neural Generation and Translation (WNGT).

[3] Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). *Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters.* arXiv preprint arXiv:2408.03314.

[4] Agarwal, R., et al. (2024). *Generalized Knowledge Distillation for Auto-regressive Sequence Models.* arXiv preprint arXiv:2306.13649.

[5] Gu, A., & Dao, T. (2023). *Mamba: Linear-Time Sequence Modeling with Selective State Spaces.* arXiv preprint arXiv:2312.00752.

[6] Lightman, H., et al. (2023). *Let's Verify Step by Step.* arXiv preprint arXiv:2305.20050.

[7] Fu, J., et al. (2026). *Revisiting On-Policy Distillation: Empirical Failure Modes and Simple Fixes.* arXiv preprint arXiv:2603.25562.

[8] NVIDIA RLHF Team. (2026). *Nemotron-Cascade 2: Multi-Domain Distillation and Cascade RL scaling.* arXiv preprint arXiv:2603.19220.

[9] Lufkin, L., Figliolia, T., Millidge, B., & Krishnamurthy, K. (2026). *Hybrid Associative Memories for Efficient Large Language Models.* arXiv preprint.

[10] Boizard, M., et al. (2025). *Universal Logit Distillation for Cross-Vocabulary Alignment.* Association for Computational Linguistics (ACL).

[11] HuggingFace H4 Team. (2026). *GOLD: General On-Policy Logit Distillation.* Open-source repository and technical report. `https://huggingface.co/docs/trl/main/en/gold`

[12] Peng, B., et al. (2025). *RWKV-7 "Goose": Generalized Delta Rule for Attention-Free Sequence Modeling.* arXiv computation and language preprint.

[13] Sun, Y., et al. (2026). *Learning to Learn at Inference: End-to-End Test-Time Training.* arXiv preprint.

[14] Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.